Convenience and Performance Charts

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1 Convenience and Performance Charts

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The following charts [Hood 2024] are intended as a handy reference for **Numerical Recipes in Fortran 77**. Each of the charts can be used as a stand-alone reference, but they are also designed to work together. Each of the charts is printed on a separate page.

The charts are organized as follows:

- The Complex Arithmetic chart lists some simple routines for manipulation of the complex number data type. Because complex numbers are not a standard data type in Fortran 77, they must be simulated by using two-element real arrays. The routines listed in this chart are the ones that I have found most useful in my own programming, and that I personally have applied to new problems.
- The Linear Algebra chart lists routines that are used to solve problems in linear algebra. The routines listed are for those most commonly encountered in practical problems.
- The Fast Fourier Transform chart lists routines for computing one-dimensional and two-dimensional Fast Fourier Transforms. The routines listed are for those most commonly encountered in practical problems.
- The Random Number Generator chart lists routines for generating pseudo-random numbers. The routines listed are for those most commonly encountered in practical problems.
- The Sorting chart lists routines for sorting data into ascending or descending order. The routines listed are for those most commonly encountered in practical problems.

Each of the charts is printed on a separate page. The charts are designed to be used as a reference, and they are not intended to be read from cover to cover.

1.1 The Charts

Algorithm	Machine Efficiency	Human Efficiency	Complexity
Cmplx_abs			
Cmplx_add			
CMPLX_DIV (IEEE)			
CMPLX_DIV (VAX)			
CMPLX_DIV (IBM)			
CMPLX_MUL (IEEE)			
Cmplx_neg			
Cmplx_sqrt			
CPLX_ABS (Hypot)			
CPLX_ABS (Safe Square Root)			
Cplx_div			
Cplx_mul			
DGELSS (LAPACK)			
DGELSS (LAPACK)			
DGELSS (LAPACK)